Hybrid Regularization of Diffusion Process for Visual Re-Ranking

Danchen Zheng, *Member, IEEE*, Jianchao Fan, *Member, IEEE*, and Min Han, *Senior Member, IEEE*

***Abstract*— To improve the retrieval result obtained from a pairwise dissimilarity, many variants of diffusion process have been applied in visual re-ranking. In the framework of diffusion process, various contextual similarities can be obtained by solving an optimization problem, and the objective function consists of a smoothness constraint and a fitting constraint. And many improvements on the smoothness constraint have been made to reveal the underlying manifold structure. However, little attention has been paid to the fitting constraint, and how to build an effective fitting constraint still remains unclear. In this article, by deeply analyzing the role of fitting constraint, we firstly propose a novel variant of diffusion process named Hybrid Regularization of Diffusion Process (HyRDP). In HyRDP, we introduce a hybrid regularization framework containing a two-part fitting constraint, and the contextual dissimilarities can be learned from either a closed-form solution or an iterative solution. Furthermore, this article indicates that the basic idea of HyRDP is closely related to the mechanism behind Generalized Mean First-passage Time (GMFPT). GMFPT denotes the mean time-steps for the state transition from one state to any one in the given state set, and is firstly introduced as the contextual dissimilarity in this article. Finally, based on the semi-supervised learning framework, an iterative re-ranking process is developed. With this approach, the relevant objects on the manifold can be iteratively retrieved and labeled within finite iterations. The proposed algorithms are validated on various challenging databases, and the experimental performances demonstrate that retrieval results obtained from different types of measures can be effectively improved by using our methods.**

***Index Terms*— Diffusion process, contextual dissimilarity, hybrid regularization, generalized mean first-passage time, iterative re-ranking process.**

I. Introduction

TRADITIONALLY, given a query object, the aim of retrieval is to find similar objects from the database according to a pairwise (dis)similarity. The basic principle is that the more similar two objects are, the smaller distance is estimated. Unfortunately, this practice often does not yield relevant results, because it suffers from the crucial limitation that the structure of underlying manifold is ignored [1], [2]. Instead, re-ranking procedure is introduced to explore contextual information from databases, and baseline results obtained from different pairwise (dis)similarities can be improved by introducing the post-processing step [3]–[8]. And the related methods have been applied in a number of different fields, such as visual tracking [9], person re-identification [10], [11], saliency detection [12], medical diagnosis [13], and phenology [14].

As discussed in [15], many of previous contextual similarities are closely related to diffusion process, and summarized in a unified iteration-based framework. These methods follow the same point that the baseline similarities are reevaluated by diffusing the similarity values through weighted graphs. Inspired by [1], [16], a new model named regularized diffusion process (RDP) is presented in recent researches [2], [17]. In RDP, the contextual similarities for diffusion process on tensor product graph are obtained by solving an optimization problem, and its objective function is taken as a regularization framework. Likewise, for many other variants of diffusion process, one can obtain a similar regularization framework, which is composed of two constraints: the smoothness constraint and the fitting constraint. Most existing algorithms focus on capturing the intrinsic manifold structure by a well-designed smoothness constraint, which have been extensively studied. However, few works have investigated the fitting constraint, and for most previous variants of diffusion process (e.g., [1], [2], [7], [17]), only a simple uniform regularization term with the square of L_2-norm is selected as the fitting constraint.

From the perspective of improving the fitting constraint, some significant questions confronting us are: (1) how to construct an effective fitting constraint by using the input baseline result; (2) what is the implications of the improved fitting constraint; (3) how to effectively solve the corresponding optimization problem. From this sense, the existing investigations of diffusion process are still insufficient, and our methods differ from previous works in three main aspects.

Firstly, we propose a novel algorithm named hybrid regularization of diffusion process (HyRDP) for learning contextual dissimilarity. Unlike other variants of diffusion process with one uniform fitting term, HyRDP contains a hybrid fitting constraint which is composed of two parts: the squared L_2-norm as a strict condition and the L_1-norm as a relaxed condition.

Manuscript received August 31, 2017; revised April 8, 2018, September 9, 2018, September 2, 2019, and December 22, 2020; accepted February 15, 2021. Date of publication March 11, 2021; date of current version March 17, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61773087, Grant 42076184, Grant 41876109, and Grant 41706195, in part by the National Key Research and Development Program of China under Grant 2017YFC1404902 and Grant 2016YFC1401007, and in part by the National High Resolution Special Research under Grant 41-Y30F07-9001-20/22. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Chunming Li. *(Corresponding author: Danchen Zheng.)*

Danchen Zheng and Min Han are with the Professional Technology Innovation Center of Distributed Control for Industrial Equipment of Liaoning Province, Dalian University of Technology, Dalian 116024, China (e-mail: dcjeong@dlut.edu.cn; minhan@dlut.edu.cn).

Jianchao Fan is with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian 116023, China (e-mail: jcfan@nmemc.org.cn).

Digital Object Identifier 10.1109/TIP.2021.3064265

In particular, we introduce two types of predefined values: the accurate default and the rough default, which are respectively used to constrain the contextual dissimilarities within different terms. So, under this structure, the contextual dissimilarities can be treated differently according to the confidence levels about the predefined values, and appropriately estimated in the learning process. Both the closed-form solution and the iterative solution are then deduced for the proposed hybrid regularization framework.

Secondly, we further explore the meaning behind the hybrid regularization framework, and demonstrate that this framework is closely related to generalized mean first-passage time (GMFPT). From the view of diffusion process, the understanding of GMFPT is extremely different from those of other variants. And it is a novel way to utilize GMFPT to represent the contextual dissimilarity. The basic point of this practice is that if the similarity information spreads rapidly through geodesic paths, the state transition between similar objects should be completed in a small number of time-steps. So, instead of evaluating contextual similarities by collectively propagating similarity values, we focus on the average time-steps for each state transition on the manifold. And as a new concept derived from mean first-passage time (MFPT), GMFPT is very easy to comprehend.

Thirdly, for the relevant object far away from the query in data space, it certainly takes lots of time-steps to achieve information propagation on the distance manifold, and then the learned contextual dissimilarity might still be very large. An iterative re-ranking process is employed to solve this problem, and it can be viewed as a semi-supervised learning method with only one category label. With the query viewed as the only labeled object at the start, the top-rankings sorted by the learned contextual dissimilarities are labeled in each iteration, and then treated as the queries together with the previously labeled ones in the next iteration. After a few iterations, the relevant objects on the manifold can be gradually searched, from the near to the far.

The rest of this article is organized as follows. We briefly review the related works in next section. The details of HyRDP and GMFPT are introduced in Section III. In Section IV, the iterative re-ranking process is further discussed, and the entire method is summarized. And in Section V, experimental results on some well-known databases are provided to illustrate the effectiveness of our methods. At last, we summarize the conclusions and the future work.

II. Related Work

Compared with the retrieval task on the basis of pairwise (dis)similarities that have been investigated for a long period of time, visual re-ranking on the basis of contextual information has only been investigated for a short period [15], [18], [19].

In [1], manifold ranking is proposed to rank the data with respect to the intrinsic manifold structure, and the basic idea is closely related to the semi-supervised learning process in [16]. Label propagation (LP) is an early variant of diffusion process [4], [20]. Let $\{1, \cdots, n-1\}$ denote unlabeled objects, and n denote the labeled one. Given $\mathbf{P}$ as the transition matrix, the information can be iteratively propagated according to a simple update rule as follows

$$\boldsymbol{\eta}_{t+1} = \mathbf{P}\boldsymbol{\eta}_t, \tag{1}$$

where $\boldsymbol{\eta}_t = [\eta_t(1), \cdots, \eta_t(n)]^\top$. $\eta_t(k)$ represents the similarity between k and n in iteration t, and $\eta_t(n)$ is fixed to 1 in the iterations. In [6], shortest path propagation (SPP) is developed as an extension of LP, and the relevant objects can be searched along the shortest paths explicitly captured.

In recent years, many different variants of diffusion process have been presented. In [21], locally constrained diffusion process (LCDP) on a k-nearest neighborhoods (kNN) graph is investigated, and the sparse data space is densified by adding synthetic points. With the self-induced smoothing kernel, a global similarity metric named self-smoothing operator (SSO) is developed in [22]. Yang *et al.* [7], [23] proposed tensor product graph (TPG) achieved by a tensor product of the original graph with itself, and a novel neighborhood structure named dominant neighborhood (DN). It is demonstrated that TPG explores the benefits of higher order relations without the price of higher computational cost. Donoser *et al.* [15] revisited different variants of diffusion process, and proposed a generic diffusion framework. Recently, RDP is proposed in [2], [17], and the essence of TPG is explained theoretically.

Enormous efforts have also been devoted to similarity fusion in the framework of diffusion process. To improve shape retrieval result, Bai *et al.* [24], [25] proposed co-transduction approach, which combines the concept of co-training and graph transduction. In [26], cross-diffusion process (CrDP) is introduced to fuse multiple metrics together. Locally constrained mixed-diffusion (LCMD) is investigated for image retrieval in [27]. In [28], [29], regularized ensemble diffusion (RED) introduces an automatic weight learning paradigm and achieves state-of-the-art performance on a variety of retrieval tasks, where the baselines are obtained from deep features based on convolutional neural networks (CNN). The framework of unified ensemble diffusion (UED), which combines the advantages of various fusion with diffusion methods, is proposed in [11] for solve the re-ranking problem in 3D shape retrieval, image retrieval and person re-identification.

The aforementioned methods are derived from the framework of diffusion process. Besides those, some re-ranking algorithms directly define a contextual similarity according to the neighborhood structures or the baseline ranking result. In [3], to improve the accuracy of bag-of-features image representation, contextual dissimilarity measure (CDM) is proposed by taking into account the local distribution of the features, and used to iteratively estimate the distance. In [30], a novel measure is presented to exploit contextual information for improving content-based image retrieval (CBIR) tasks. In [19], a novel feature vector named sparse contextual activation (SCA) is proposed by considering the pairwise distance in the contextual space, and the re-ranking task can be simply accomplished by vector comparison. Re-ranking with kNN [31], mutual kNN (mkNN) [32] and reciprocal kNN [33] are also investigated in previous studies.

First-passage time (FPT) and MFPT are important concepts of stochastic process, which have been applied in many research areas [34], [35]. As a variant of MFPT, the mean absorbed times based on an absorbing Markov chain is employed to solve the task of saliency detection [36], [37]. As a general form of MFPT, GMFPT is firstly proposed for fusing similarities in our early work [38], but the principle behind it has not yet been explored. In this article, by analysis of the mechanism behind GMFPT, it is indicated that GMFPT is closely related to the proposed hybrid regularization framework.

In many cases, the learned contextual dissimilarity might be a non-metric distance, which does not satisfy the triangle inequality. It is worth noting that this violation of the triangle inequality often plays a great role in retrieval tasks. As discussed in [39], [40], the strong triangle inequality can be replaced by a weaker form: the relaxed triangle inequality, and the symmetry property does not always hold either. In this article, for the proposed contextual dissimilarity, the relaxed triangle inequality can be easily proved, and we do not provide the proofs here.

III. Contextual Dissimilarity Based on Hybrid Regularization of Diffusion Process

Firstly, the related notations are given in a preliminary summary in Section III-A. Then, HyRDP and the corresponding solutions are provided in Section III-B and Section III-C. And GMFPT is introduced and discussed in Section III-D.

A. Preliminary

Let $S = \{1, 2, \cdots, n\}$ be a collection of objects, and $\mathbf{W} = \left[w_{ij}\right]_{n\times n}$ be a symmetric weight matrix derived from the input baseline result with noisy similarities removed. The construction of $\mathbf{W}$ is discussed in Section IV-B. We build a weighted graph $\mathcal{G}$ with S as the vertices and $\{w_{ij} : w_{ij} \in \mathbf{W}\}$ as the edges, and thus $w_{ij} = 0$ means there is no connection between i and j. Then, the transition matrix is obtained as $\mathbf{P} = \mathbf{D}^{-1}\mathbf{W} = \left[p_{ij}\right]_{n\times n}$, where $\mathbf{D}$ is a $n \times n$ diagonal matrix with $d_{ii} = \sum_{k=1}^{n} w_{ik}$.

S contains two subsets: L with l labeled objects and U with u unlabeled ones. U is further divided into two subsets: A with a objects and B with b objects, where $A = \{i : \sum_{k\in L} w_{ik} > 0, i \in U\}$, $B = \{j : \sum_{k\in L} w_{jk} = 0, j \in U\}$, and $U = A + B$. We rearrange all the states so that unlabeled states are placed before labeled ones, and the states from A are listed before those from B. Then, we have $U = \{1, \cdots, u\}$, $L = \{u + 1, \cdots, n\}$, $A = \{1, \cdots, a\}$ and $B = \{a+1, \cdots, u\}$. Then, $\mathbf{W}$, $\mathbf{D}$ and $\mathbf{P}$ can be partitioned into blocks:

1) : $\mathbf{W}$ is partitioned into 4 blocks as

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}_{UU} & \mathbf{W}_{UL} \\ \mathbf{W}_{LU} & \mathbf{W}_{LL} \end{bmatrix}, \tag{2}$$

where $\mathbf{W}_{UU} = [w_{ii'}]_{u\times u}$, $\mathbf{W}_{UL} = \left[w_{ij'}\right]_{u\times l}$, $\mathbf{W}_{LU} = \left[w_{ji'}\right]_{l\times u}$, and $\mathbf{W}_{LL} = \left[w_{jj'}\right]_{l\times l}$.

2) : $\mathbf{D}$ is partitioned into blocks as

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_{U} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_{L} \end{bmatrix}, \tag{3}$$

where $\mathbf{D}_U$ is a $u\times u$ diagonal matrix, and $\mathbf{D}_L$ is a $l\times l$ diagonal matrix. And $\mathbf{D}_U$ can be written as blocks

$$\mathbf{D}_U = \begin{bmatrix} \mathbf{D}_{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_{B} \end{bmatrix}, \tag{4}$$

where $\mathbf{D}_A$ is a $a \times a$ diagonal matrix, and $\mathbf{D}_B$ is a $b \times b$ diagonal matrix.

3) : $\mathbf{P}$ is partitioned into 4 blocks as

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_{UU} & \mathbf{P}_{UL} \\ \mathbf{P}_{LU} & \mathbf{P}_{LL} \end{bmatrix}, \tag{5}$$

where $\mathbf{P}_{UU} = [p_{ii'}]_{u\times u}$, $\mathbf{P}_{UL} = \left[p_{ij'}\right]_{u\times l}$, $\mathbf{P}_{LU} = \left[p_{ji'}\right]_{l\times u}$, and $\mathbf{P}_{LL} = \left[p_{jj'}\right]_{l\times l}$.

We define two $u \times u$ diagonal matrices: $\mathbf{H}_U$ with $h_{ii} = \sum_{i'=1}^{u} w_{ii'}$ and $\mathbf{E}_U$ with $e_{ii} = \sum_{j'=u+1}^{n} w_{ij'}$, where $w_{ii'} \in \mathbf{W}_{UU}$ and $w_{ij'} \in \mathbf{W}_{UL}$. $\mathbf{E}_U$ can be written as blocks

$$\mathbf{E}_U = \begin{bmatrix} \mathbf{E}_{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, \tag{6}$$

where $\mathbf{E}_A$ is a $a \times a$ diagonal matrix, and $\mathbf{D}_U = \mathbf{H}_U + \mathbf{E}_U$.

In this article, $\mathbf{I}$ denotes the identity matrix, and $\boldsymbol{r}_k$ is used to represent a k-dimensional column vector with each element having the value $r \in \mathbb{N}^+$. In some cases, we use $\mathbf{1}$ instead of $\mathbf{1}_k$ for convenience. $\|\cdot\|_2$ and $\|\cdot\|_1$ denote L_2-norm and L_1-norm, respectively.

In the graph $\mathcal{G}$, $i \sim j$ denotes that $i \in S$ and $j \in S$ are connected. And $i \sim L$ denotes that there is at least one node $j \in L$ that satisfies $i \sim j$ for $i \notin L$. Unless specified otherwise, we suppose that $i \sim L$ for $\forall i \in U$.

B. Hybrid Regularization Framework

We use ψ_{iL} to represent the contextual dissimilarity from $i \in S$ to L, and always have $\psi_{kL} = 0$ for $k \in L$. Then our aim is to obtain $\boldsymbol{\psi}_L = [\psi_{1L}, \cdots, \psi_{uL}]^\top$ with the optimization problem defined as

$$\begin{aligned} &\min_{\boldsymbol{\psi}_L} \mathcal{J}_h\left(\boldsymbol{\psi}_L\right) \\ &\text{s.t.} \quad \psi_{iL} \geq 0, \quad i = 1, \cdots, u, \end{aligned} \tag{7}$$

and we have $\mathcal{J}_h\left(\boldsymbol{\psi}_L\right) = \mathcal{S}_h\left(\boldsymbol{\psi}_L\right) + \mathcal{F}_h\left(\boldsymbol{\psi}_L\right)$, where $\mathcal{S}_h$ and $\mathcal{F}_h$ denote the smoothness constraint and the fitting constraint respectively. $\mathcal{S}_h$ is defined as

$$\begin{aligned} \mathcal{S}_h &= \frac{1}{2}\sum_{i,j=1}^{u} w_{ij}\left(\frac{\psi_{iL}}{\sqrt{h_{ii}}} - \frac{\psi_{jL}}{\sqrt{h_{jj}}}\right)^2 \\ &= \frac{1}{2}\boldsymbol{\psi}_L^\top\left(\mathbf{I} - \mathbf{H}_U^{-\frac{1}{2}}\mathbf{W}_{UU}\mathbf{H}_U^{-\frac{1}{2}}\right)\boldsymbol{\psi}_L. \end{aligned} \tag{8}$$

And $\mathcal{F}_h$ is defined as

$$\begin{aligned} \mathcal{F}_h &= \frac{1}{2}\alpha\sum_{i=1}^{a}\frac{e_{ii}}{d_{ii}}\left(\psi_{iL} - \frac{d_{ii}}{e_{ii}}\right)^2 + \beta\sum_{j=a+1}^{u}\left|\psi_{jL} - \xi\right| \\ &= \frac{1}{2}\alpha\left\|\mathbf{D}_A^{-\frac{1}{2}}\mathbf{E}_A^{\frac{1}{2}}\boldsymbol{\psi}_{AL} - \mathbf{D}_A^{\frac{1}{2}}\mathbf{E}_A^{-\frac{1}{2}}\mathbf{1}\right\|_2^2 + \beta\left\|\boldsymbol{\psi}_{BL} - \boldsymbol{\xi}_b\right\|_1, \end{aligned} \tag{9}$$

where $\boldsymbol{\psi}_{AL} \in \mathbb{R}^a$, $\boldsymbol{\psi}_{BL} \in \mathbb{R}^b$, and $\boldsymbol{\psi}_L = \left[\boldsymbol{\psi}_{AL}^\top, \boldsymbol{\psi}_{BL}^\top\right]^\top$. $\alpha > 0$ and $\beta > 0$ are the regularization hyper-parameters.

Parameter ξ is the rough predefined value, which is discussed later in this subsection.

As can be seen from Eq. 8 and Eq. 9, $\mathcal{S}_h$ and $\mathcal{F}_h$ are both derived from $\mathbf{W}$. For the smoothness constraint, $\mathbf{W}$ determines the relationship between different contextual dissimilarities expected to be learned. In previous works, the smoothness term is introduced to restrict the classifying functions of two tuples [1], [16] or the relations between four tuples [2], [17]. In contrast, we focus on a special form of three tuples: two elements and one set. In $\mathcal{S}_h$, each condition between two normalized contextual dissimilarities is generated with a element from $\mathbf{W}_{UU}$. And it means that if i and j share a large w_{ij}, then the normalized ψ_{iL} and ψ_{jL} should be similar, where $i, j \notin L$.

For the fitting constraint $\mathcal{F}_h$, the inherent meaning is more complicated. In previous works [2], [16], [17], the guideline for fitting constraint is that the learned similarities are not expected to change too much from the predefined values, and then a squared L_2-norm is introduced based on either an identity matrix or the similarity matrix. However, our proposed fitting constraint is extremely different from the previous ones, and includes two implications. Firstly, the predefined values are generated as precisely as possible, and thus two types of predefined values are defined respectively. Secondly, different condition terms are further constructed according to the confidence levels about the predefined values.

In the first place, $\mathcal{F}_h$ involves two types of predefined values: the accurate default and the rough default. On the one hand, since each element in A is a neighbor of one or several elements in L, the relationship between $\forall i \in A$ and L can be precisely described by the similarities in $\mathbf{W}$. So, the accurate default for ψ_{iL} is set to $\frac{d_{ii}}{e_{ii}}$, which is inversely proportional to $\sum_{j=u+1}^{n} p_{ij}$. On the other hand, there is no direct connection from $\forall j \in B$ to L, the predefined value for $\psi_{jL} \in \boldsymbol{\psi}_{BL}$ certainly should be much larger than any accurate default. Also because there is no connection between $\forall j \in B$ and L, their relationship cannot be accurately described, which means the corresponding predefined values cannot be precisely determined. So, these uncertain predefined values are treated equally, and denoted by the rough default ξ, where $\xi \gg \max_{i \in A} \frac{d_{ii}}{e_{ii}}$. As the elements expected to be learned are bounded, we set a very large positive value to ξ, and have $\xi > \max\left(\boldsymbol{\psi}_{BL}\right)$ for the convenience of solving the optimization problem discussed later.

In the second place, two different terms, the squared L_2-norm and the L_1-norm, are further constructed based on the accurate default and the rough default, respectively. For the first term, ψ_{iL} for $i \in A$ should be as close as possible to the accurate default $\frac{d_{ii}}{e_{ii}}$, so the quadratic term is constructed as a strict condition. And an extra addition weight $\frac{e_{ii}}{d_{ii}} = \sum_{j=u+1}^{n} p_{ij}$ is attached to this term. For the second term, we are uncertain about the rough defaults that are used to restrict the rest contextual dissimilarities. So, compared with the strict condition, a larger difference between ξ_j and $\psi_{jL} \in \boldsymbol{\psi}_{BL}$ should be allowed to exist. In this sense, the L_1-norm as the relaxed condition is generated.

In the hybrid regularization framework, α and β are introduced to balance the smoothness constraint and two terms of the fitting constraint. It is worth noting that the regularization terms in HyRDP are different from those in ElasticNet regularization. For the ElasticNet, all the parameters are treated equally, and each one is limited by both L_2-norm and L_1-norm. However, for HyRDP, $\boldsymbol{\psi}_L$ is divided into two groups that are respectively associated with the squared L_2-norm and the L_1-norm. In another sense, the point of introducing L_1-norm in ElasticNet is to achieve a sparse solution, and completely different from that in HyRDP.

C. Solution of the Proposed Framework

In general, with a L_1-norm term, the optimal solution requires complex operations. However, since we already have $\xi > \max\left(\boldsymbol{\psi}_{BL}\right)$, the fitting constraint $\mathcal{F}_h$ in Eq. 9 can be rewritten as

$$\mathcal{F}_h = \frac{1}{2}\alpha \left\| \mathbf{D}_A^{-\frac{1}{2}} \mathbf{E}_A^{\frac{1}{2}} \boldsymbol{\psi}_{AL} - \mathbf{D}_A^{\frac{1}{2}} \mathbf{E}_A^{-\frac{1}{2}} \mathbf{1} \right\|_2^2 - \beta \boldsymbol{\psi}_{BL}^{\top} \mathbf{1} + const. \tag{10}$$

And then, Eq. 7 can be simplified as a non-constrained optimization problem, and we have

$$\boldsymbol{\psi}_L^* = \arg\min_{\boldsymbol{\psi}_L} \mathcal{J}_h\left(\boldsymbol{\psi}_L\right). \tag{11}$$

By taking the partial derivative of $\mathcal{J}_h\left(\boldsymbol{\psi}_L\right)$ with $\boldsymbol{\psi}_L$, we have

$$\begin{aligned}\frac{\partial \mathcal{J}_h}{\partial \boldsymbol{\psi}_L} = &\left(\mathbf{I} - \mathbf{H}_U^{-\frac{1}{2}} \mathbf{W}_{UU} \mathbf{H}_U^{-\frac{1}{2}}\right) \boldsymbol{\psi}_L \\ &+ \alpha \left(\mathbf{D}_U^{-1} \mathbf{E}_U \boldsymbol{\psi}_L - \begin{bmatrix} \mathbf{1}_a \\ \mathbf{0}_b \end{bmatrix}\right) - \beta \begin{bmatrix} \mathbf{0}_a \\ \mathbf{1}_b \end{bmatrix}. \end{aligned} \tag{12}$$

By setting Eq. 12 to zero, we obtain

$$\mathbf{H}_U^{-\frac{1}{2}} \left(\Lambda - \mathbf{W}_{UU}\right) \mathbf{H}_U^{-\frac{1}{2}} \boldsymbol{\psi}_L^* = \begin{bmatrix} \alpha \mathbf{1}_a \\ \beta \mathbf{1}_b \end{bmatrix}, \tag{13}$$

where $\Lambda = \mathbf{H}_U + \alpha \mathbf{H}_U \mathbf{D}_U^{-1} \mathbf{E}_U$ is a $u \times u$ diagonal matrix. According to APPENDIX A, $\rho\left(\Lambda^{-1} \mathbf{W}_{UU}\right) < 1$ can be obtained, and the coefficient matrix in Eq. 13 is invertible, where $\rho(\cdot)$ denotes the spectral radius. Thus, we have the closed-form solution as

$$\boldsymbol{\psi}_L^* = f(\Lambda) \begin{bmatrix} \alpha \mathbf{1}_a \\ \beta \mathbf{1}_b \end{bmatrix}, \tag{14}$$

where $f(\Lambda) = \mathbf{H}_U^{\frac{1}{2}} \left(\mathbf{I} - \Lambda^{-1} \mathbf{W}_{UU}\right)^{-1} \Lambda^{-1} \mathbf{H}_U^{\frac{1}{2}}$ is a matrix function.

To avoid the matrix inversion, the iterative solution can be obtained from Eq. 14 if $\left(\mathbf{I} - \Lambda^{-1} \mathbf{W}_{UU}\right)^{-1}$ is written as the sum of matrix power series. Given a column vector $\boldsymbol{y} = \Lambda^{-1} \mathbf{H}_U^{\frac{1}{2}} \left[\alpha \mathbf{1}_a^{\top}, \beta \mathbf{1}_b^{\top}\right]^{\top}$, we define $\boldsymbol{x}^1 = \boldsymbol{y}$ as the initial value and have

$$\boldsymbol{x}^{\tau+1} = \Lambda^{-1} \mathbf{W}_{UU} \boldsymbol{x}^{\tau} + \boldsymbol{y}, \tag{15}$$

whose time complexity is $O\left(u^2\right)$ because of the matrix-vector multiplication, where u denotes the number of unlabeled objects. Considering the summation of top m items in the series, the solution can be obtained as

$$\boldsymbol{\psi}_L^* = \mathbf{H}_U^{\frac{1}{2}} \boldsymbol{x}^m, \tag{16}$$

and thus we have the time complexity for the iterative solution is $O\left(mu^2\right)$. According to Eq. 16, the iteration number directly depends on the convergence rate when we need the accurate numerical solution. But practically, during the re-ranking process, we are more concerned with a small number of top ranks, which might remain unchanged long before the iteration converges, than the final learned dissimilarities. In our experiments, the value of m generally increases with the increasing of u or the number of top ranks we care about.

D. Generalized Mean First-Passage Time

With the transition matrix $\mathbf{P}$, we define $\{X_k : k \geq 0\}$ as a Markov chain, which can be treated as a random walk on the graph $\mathcal{G}$. Given a random walk starting from state i, GMFPT μ_{iL} denotes the expected number of time-steps for reaching any state in L for the first time. And then we have

$$\mu_{iL} = \mathrm{E}\left(T_L | X_0 = i\right) = \sum_{k=1}^{\infty} k \mathrm{Pr}\left(T_L = k | X_0 = i\right), \tag{17}$$

where $T_L = \min\{k > 0 : X_k \in L\}$. In particular, if L only contains one state j, we have MFPT μ_{ij} defined as follows

$$\mu_{ij} = \mathrm{E}\left(T_j | X_0 = i\right) = \sum_{k=1}^{\infty} k \mathrm{Pr}\left(T_j = k | X_0 = i\right). \tag{18}$$

As given in APPENDIX B, the closed-form solution of GMFPTs can be obtained as

$$\boldsymbol{\mu}_{SL} = \left(\mathbf{I} - \begin{bmatrix} \mathbf{P}_{UU} & \mathbf{0} \\ \mathbf{P}_{LU} & \mathbf{0} \end{bmatrix}\right)^{-1} \mathbf{1}_n. \tag{19}$$

where $\boldsymbol{\mu}_{SL} = [\mu_{1L}, \cdots, \mu_{nL}]^\top \in \mathbb{R}^n$. As mentioned before, we only concentrate on the contextual dissimilarity from each unlabeled object to the labeled ones. So, we have

$$\boldsymbol{\mu}_L = \left(\mathbf{I} - \mathbf{P}_{UU}\right)^{-1} \mathbf{1}_u, \tag{20}$$

where $\boldsymbol{\mu}_L = [\mu_{1L}, \cdots, \mu_{uL}]^\top$.

Compared to other variants of diffusion process, such as LP [4], GMFPT achieves better performance on capturing the intrinsic manifold structure. According to Eq. 1 of LP, $\eta_t(n)$ for n is fixed to 1, and always larger than $\eta_t(k)$ for $\forall k \neq n$. So, for a node directly connected to n with a large weight, it could receive more similarity information than any node without a direct connection to n or with a small connection to n does. Thus, the role of $p_{in} \in \mathbf{P}$ is exaggerated during the propagation process, and the similarity information cannot fairly spread. On the other hand, GMFPT cares about the time-steps for the state transition, and all the connections between different objects are treated equally. And the state transition between two similar objects could be achieved in a few time-steps via relevant states on the shortest paths.

Moreover, we examine the optimization problem behind GMFPT, and find $\boldsymbol{\mu}_L$ can be obtained by solving the following optimization problem

$$\begin{aligned} &\min_{\boldsymbol{\mu}_L} \mathcal{J}_g\left(\boldsymbol{\mu}_L\right) \\ &\text{s.t.} \quad \mu_{iL} \geq 0, \ i = 1, \cdots, u, \end{aligned} \tag{21}$$

where $\mathcal{J}_g\left(\boldsymbol{\mu}_L\right) = \mathcal{S}_g\left(\boldsymbol{\mu}_L\right) + \mathcal{F}_g\left(\boldsymbol{\mu}_L\right)$. The smoothness constraint $\mathcal{S}_g$ is defined as

$$\begin{aligned} \mathcal{S}_g &= \frac{1}{2} \sum_{i,j=1}^{u} w_{ij} \left(\mu_{iL} - \mu_{jL}\right)^2 \\ &= \frac{1}{2} \boldsymbol{\mu}_L^\top \left(\mathbf{H}_U - \mathbf{W}_{UU}\right) \boldsymbol{\mu}_L. \end{aligned} \tag{22}$$

And the fitting constraint $\mathcal{F}_g$ is defined as

$$\begin{aligned} \mathcal{F}_g &= \frac{1}{2} \sum_{i=1}^{a} e_{ii} \left(\mu_{iL} - \frac{d_{ii}}{e_{ii}}\right)^2 + \sum_{j=a+1}^{u} d_{jj} \left|\mu_{jL} - \xi\right| \\ &= \frac{1}{2} \left\| \mathbf{E}_A^{\frac{1}{2}} \boldsymbol{\mu}_{AL} - \mathbf{D}_A \mathbf{E}_A^{-\frac{1}{2}} \mathbf{1} \right\|_2^2 + \left\| \mathbf{D}_B \left(\boldsymbol{\mu}_{BL} - \boldsymbol{\xi}_b\right) \right\|_1, \end{aligned} \tag{23}$$

where $\boldsymbol{\mu}_{AL} \in \mathbb{R}^a$, $\boldsymbol{\mu}_{BL} \in \mathbb{R}^b$, $\boldsymbol{\mu}_L = \left[\boldsymbol{\mu}_{AL}^\top, \boldsymbol{\mu}_{BL}^\top\right]^\top$, and $\xi > \max\left(\boldsymbol{\mu}_{BL}\right)$. So $\mathcal{F}_g$ can be rewritten as

$$\mathcal{F}_g = \frac{1}{2} \left\| \mathbf{E}_A^{\frac{1}{2}} \boldsymbol{\mu}_{AL} - \mathbf{D}_A \mathbf{E}_A^{-\frac{1}{2}} \mathbf{1} \right\|_2^2 - \left(\mathbf{D}_B \boldsymbol{\mu}_{BL}\right)^\top \mathbf{1} + const. \tag{24}$$

Similar to HyRDP, GMFPTs can be directly obtained from

$$\boldsymbol{\mu}_L^* = \arg\min_{\boldsymbol{\mu}_L} \mathcal{J}_g\left(\boldsymbol{\mu}_L\right), \tag{25}$$

and we have

$$\frac{\partial \mathcal{J}_g}{\partial \boldsymbol{\mu}_L} = \left(\mathbf{H}_U - \mathbf{W}_{UU} + \mathbf{E}_U\right) \boldsymbol{\mu}_L - \mathbf{D}_U \mathbf{1}_u. \tag{26}$$

Then the closed-form solution in Eq. 20 is obtained by setting Eq. 26 to zero. And it is obvious that the time complexity of GMFPT is the same as that of HyRDP.

As observed from $\mathcal{J}_h$ and $\mathcal{J}_g$, the hybrid regularization frameworks of HyRDP and GMFPT are essentially the same. Specifically, for the smoothness constraint, $\mathcal{S}_h$ can be obtained from $\mathcal{S}_g$ by adding a normalized coefficient to each variable. For the fitting constraint, $\mathcal{F}_h$ and $\mathcal{F}_g$ share the same hybrid regularization structure, and the former one can be derived from the latter one by adjusting the weights and introducing the hyper-parameters. So, GMFPT can be taken as a special form of HyRDP with the hyper-parameters fixed to 1.

IV. Re-Ranking Method Based on Proposed Model

Based on the proposed model, an iterative re-ranking process in the semi-supervised learning framework is introduced and discussed in Section IV-A. The entire re-ranking method with the pseudocode is given and discussed in Section IV-B.

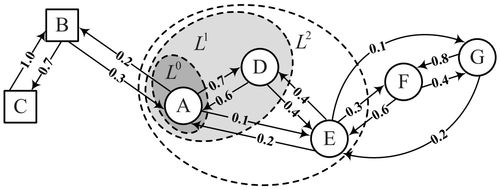


Fig. 1. An example of the iterative re-ranking process based on GMFPT.

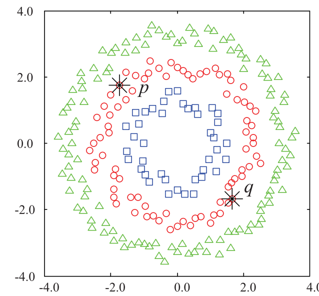


Fig. 2. Three-circles synthetic dataset.

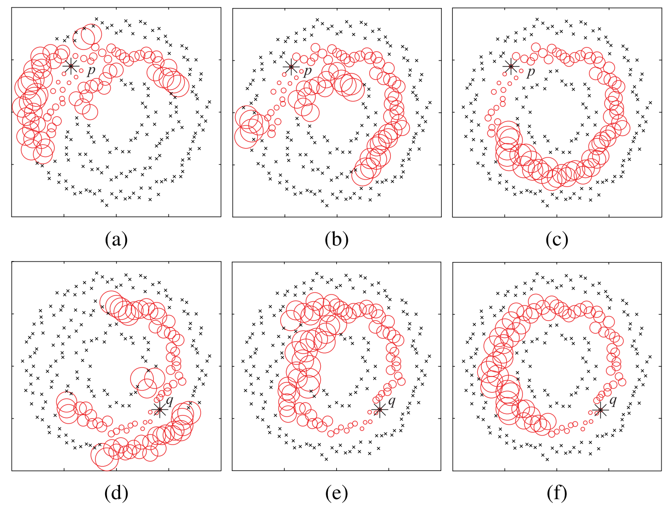


Fig. 3. Re-ranking results of Fig. 2 based on different approaches. p and q are selected as the queries in (a, b, c) and (d, e, f), respectively. (a, d) Performance of LP; (b, e) Performance of GMFPT; (c, f) Performance of the iterative re-ranking process based on GMFPT. The top 80 retrieved samples are shown as rings with sizes increasing with rankings.

A. Iterative Re-Ranking Process

From the view of diffusion process, the similarity information cannot spread to the relevant objects very far away from the query in a short time, because it has to travel through a large number of nodes between them on the geodesic paths. Therefore, these relevant objects certainly rank poorly according to the contextual dissimilarities.

In the semi-supervised learning framework, we introduce an iterative re-ranking process to address this concern. At the start, we treat the query and the others as a labeled object and unlabeled ones, respectively. In each iteration, after the contextual dissimilarities are evaluated, we label a few top-ranked samples that will be taken as the queries in the next iteration, together with the previously labeled ones. So after iterations, the relevant objects can be gradually searched and labeled, and the structure of the underlying manifold is explicitly captured.

Let L^t and U^t respectively denote labeled and unlabeled ones in iteration t. Given $L^0 = \{n\}$ and $U^0 = \{1, \cdots, n-1\}$ as the initial values, each iteration consists of two procedures: evaluating $\boldsymbol{\psi}_{L^t}$ and updating $L^{t+1} = L^t + \Delta^t$, where $\Delta^t = \arg\min_{\lambda, i\in U^t} (\psi_{iL^t})$. $\arg\min_\lambda(\cdot)$ returns the λ minimum elements, and λ is the step-size. After T iterations, the re-ranking result can be obtained according to the order in which the objects are labeled and added to L^T. A similar process for GMFPT can be obtained by replacing $\boldsymbol{\psi}_{L^t}$ with $\boldsymbol{\mu}_{L^t}$.

Fig. 1 provides an example to explain the iterative re-ranking process with GMFPT. Let $\{\mathrm{A}, \mathrm{B}, \cdots, \mathrm{G}\}$ denote the collection composed of two categories, and the elements from different categories are represented by circles and squares respectively. The probabilities marked on arrow lines are inversely related to pairwise distances. With dis_{XY} representing the pairwise distance between X and Y, we have $dis_{\mathrm{BA}} < dis_{\mathrm{EA}}$, $dis_{\mathrm{BA}} < dis_{\mathrm{FA}}$ and $dis_{\mathrm{BA}} < dis_{\mathrm{GA}}$, which is obviously incorrect. With A as the query, we have $\mu_{\mathrm{EA}} < \mu_{\mathrm{BA}}$, but the other rankings are still unsatisfying, e.g. $\mu_{\mathrm{FA}} > \mu_{\mathrm{BA}}$ and $\mu_{\mathrm{GA}} > \mu_{\mathrm{BA}}$. Whereas, with $\lambda = 1$, a desirable re-ranking result is obtained in several iterations. As shown in Fig. 1, given $L^0 = \{\mathrm{A}\}$ at the start, $L^1 = \{\mathrm{A}, \mathrm{D}\}$ and $L^2 = \{\mathrm{A}, \mathrm{D}, \mathrm{E}\}$ can be obtained from $\boldsymbol{\mu}_{L^0}$ and $\boldsymbol{\mu}_{L^1}$, respectively. After 2 iterations, we have $\mu_{\mathrm{F}L^2} < \mu_{\mathrm{B}L^2}$, $\mu_{\mathrm{G}L^2} < \mu_{\mathrm{B}L^2}$, and thus the correct result $\{\mathrm{A}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}, \mathrm{B}, \mathrm{C}\}$ is obtained.

In Fig. 2, another example based on a synthetic dataset is provided. This dataset with noises are sampled from three concentric circles with different radiuses, and it contains 40, 80 and 120 elements expressed as squares, rings and triangles, respectively. p and q are two samples randomly selected from the middle circle. Our aim is to retrieve all the elements from the same pattern with either p or q as the query. Given L_2-norm as the baseline, the results for three different algorithms are shown in Fig. 3. In these drawings, each element from the top 80 is marked with ring, whose radius increases with the ranking value. And the rests are labeled with crosses. In Fig. 3a and Fig. 3d, many samples from the other two circles are wrongly labeled by LP. Fig. 3b and Fig. 3e provide better re-ranking performance obtained by GMFPT, but a few samples from the other patterns are still incorrectly labeled. According to the iterative re-ranking process, all the relevant sample are effectively retrieved and correctly labeled with $\lambda = 5$, as shown in Fig. 3c and Fig. 3f. More experiments about this synthetic dataset are given in Section V-A.

As previously discussed, in the iterative re-ranking process, only λ top-ranked samples are labeled and added to L^t in iteration t. We can ignore the contextual dissimilarities for the samples far away from L^t, because they are certainly much greater than the top λ ranks. So, in each iteration, we are only concerned with a small number of objects within the small area adjacent to L^t, and a time-variant S^t based on $\mathbf{W}$ is defined as

$$S^t = L^t + A^t + B^t, \tag{27}$$

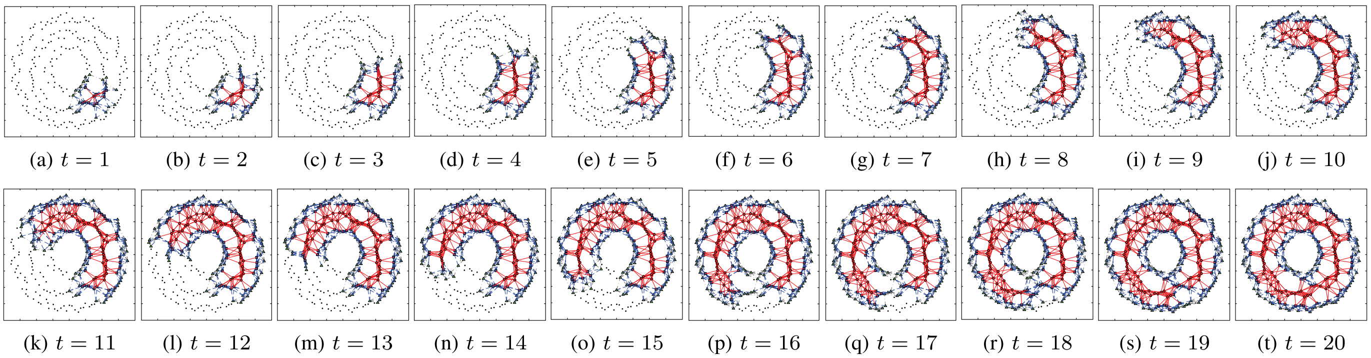


Fig. 4. Re-ranking results of Fig. 2 in 20 iterations based on the time-variant $\mathcal{G}^t$.

where $A^t = \{i : \sum_{k \in L^t} w_{ik} > 0, i \notin L^t\}$, $B^t = \{j : \sum_{k \in A^t} w_{jk} > 0, j \notin L^t + A^t\}$, and $U^t = A^t + B^t$. Then, a time-variant $\mathcal{G}^t$ is generated based on S^t, and the corresponding contextual dissimilarities can be further obtained.

An example about the re-ranking process with a time-variant $\mathcal{G}^t$ is provided in Fig. 4. With the dataset and the query q in Fig. 2, the re-ranking results for 20 iterations with $\lambda = 5$ are given. The samples from L^t, A^t and B^t are expressed as rings, squares and triangles respectively, and the rests are marked with dots. The solid lines denote the edges inside L^t, and the ones connecting L^t and A^t. The dashed lines represent the edges inside A^t, and the ones connecting A^t and B^t. The dotted lines are taken as the edges inside B^t. As can be observed, the underlying manifold structure are gradually captured within 20 iterations.

B. Re-Ranking Method

As discussed in previous works [23], [41], the pairwise (dis)similarities are more accurate for close neighbors, and it is nature to remove noisy similarities by constraining the relation from each element to its neighbors. In this article, to remove noisy similarities and keep meaningful connections, $\mathbf{W} = \left[w_{ij}\right]_{n\times n}$ is constructed based on reciprocal kNN as

$$w_{ij} = \begin{cases} sim_{ij} & i \in \mathcal{N}_s(j), j \in \mathcal{N}_s(i) \\ 0 & \text{others,} \end{cases} \tag{28}$$

where sim_{ij} denotes the input similarity, and $\mathcal{N}_s(k)$ denotes K_s nearest neighbors of k determined by sim_{ij}. In many cases, the baseline result is derived from a distance measure, rather than a similarity measure. Then, the similarity sim_{ij} can be achieved by

$$sim_{ij} = \exp\left(-\frac{dis_{ij}^2}{\sigma_{ij}^2}\right), \tag{29}$$

and σ_{ij} is obtained as follows

$$\sigma_{ij} = \frac{\gamma}{2K_d}\left(\sum_{k\in\mathcal{N}_d(i)} dis_{ik} + \sum_{k\in\mathcal{N}_d(j)} dis_{jk}\right), \tag{30}$$

where dis_{ij} denotes the input distance, and $\mathcal{N}_d(k)$ denotes K_d nearest neighbors of k determined by dis_{ij}. The choice of the parameters: γ, K_d and K_s, is discussed in detail in Section V-F.

In Section III-A, it is assumed that $i \sim L$ for $\forall i \in U$ is always true. But in general, $\mathcal{G}$ is likely to be composed of connected components. And if i belongs to a component, to which any element from L does not belong, then $i \sim L$ certainly does not satisfied. According to the definition of GMFPT, we have $\mu_{iL} \to \infty$, which is meaningless for re-ranking. And the same goes for HyRDP. During the iteration process, some temporary connections between different connected components are created to address this problem.

A transition matrix $\tilde{\mathbf{P}} = \left[\tilde{p}_{ij}\right]_{n\times n}$ is defined based on a fully connected graph with the set S of vertices together with the set $\{sim_{ij} : 1 \le i, j \le n\}$ of edges. In iteration t, the elements in L^t belong to a certain number of connected components, whose vertices are contained in a collection G^t. After removing these connected components from $\mathcal{G}$, we have K^t connected components left, and the vertices from the kth one is represented by the set C_k, where $1 \le k \le K^t$. And we define $\Gamma^t = \arg\max_{\lambda, 1\le k\le K^t} \left(q_k^t\right)$ for selecting λ connected components from $\{C_k : 1 \le k \le K^t\}$, where $q_k^t = \max\left(\{\tilde{p}_{ij} | i \in C_k, j \in G^t\}\right)$. Then, $\bar{\mathbf{W}} = \left[\bar{w}_{ij}\right]_{n\times n}$ with temporary connections is defined as follows

$$\bar{w}_{ij} = \begin{cases} sim_{ij} & (i, j) \in \mathbf{\Omega},\ \ (j, i) \in \mathbf{\Omega} \\ w_{ij} & \text{others,} \end{cases} \tag{31}$$

where $\mathbf{\Omega} = \{(i, j) \,|\, (i, j) = \arg\max_{x\in C_k, y\in G^t} \left(\tilde{p}_{xy}\right), k \in \Gamma^t\}$.

Given N_Q as the number of objects expected to be retrieved, the iteration number T for the iterative re-ranking process is determined by λ. The pseudocode of the iterative re-ranking process based on HyRDP (I-HyRDP) is given in Algorithm 1. And the iterative re-ranking process based on GMFPT (I-GMFPT) can be obtained by replacing $\boldsymbol{\psi}_{L^t}$ with $\boldsymbol{\mu}_{L^t}$.

Moreover, with a large N_Q but a small number of similar objects in the dataset, $\mathbf{I} - \Lambda^{-1}\mathbf{W}_{UU}$ or $\mathbf{I} - \mathbf{P}_{UU}$ become a near singular matrix before we have all the N_Q samples retrieved. To solve this concern, after a certain number of iterations, we could rank the unlabeled samples according to the input distances, and add them to the end of L^t in order.

Algorithm 1 I-HyRDP

Input: $\mathbf{W} = [w_{ij}]_{n\times n}$, $L^0 = \{n\}$, and $t = 0$.
Output: L^T

while $\#L^t < N_Q$ **do**
  if $\#L^t = \#G^t$ **then**
    create $\bar{\mathbf{W}}$ by Eq. 31;
    create S^t by Eq. 27 with $\bar{\mathbf{W}}$;
  else
    create S^t by Eq. 27 with $\mathbf{W}$;
  if $\#U^t \le \lambda$ **then**
    $L^{t+1} = L^t + U^t$;
  else
    calculate $\boldsymbol{\psi}_{L^t}$ by Eq. 14 or Eq. 16;
    $L^{t+1} = L^t + \Delta^t$, $\Delta^t = \arg\min_{i\in U^t}{}_\lambda\,(\psi_{iL^t})$;
  $t = t + 1$.

V. Experimental Results

To evaluate the validity of our methods, some experiments are conducted on different datasets. The baseline results are obtained on different visual features, including hand-crafted features and CNN-based features. With each identical baseline, our methods are compared with the state-of-the-art methods: LP [4], mkNN [32], TPG + DN [23], GDP [15], SCA [19], and RDP [17]. It should be noted that, to provide fair comparisons between the proposed methods and the others, we either use published results or run source codes released by the authors. The effects of the parameters are discussed in Section V-F, and the time complexity analysis is analyzed in Section V-G.

In the following experiments, *retrieval score* is selected to measure the re-ranking performances of different methods. During each retrieval process, one object is selected as the query, the number of correct results in the top K ranks is counted, where K denotes the number of objects belonging to the same category of the query. After each object in the dataset has been selected as the query, we calculate the ratio of the total correct results over the total relevant samples, which is defined as the retrieval score. In the following experiments, N_Q as the number of samples expected to be retrieved is equal to the number of objects from each category. Meanwhile, we also plot the precision-recall (P-R) curves to illustrate the detail comparisons of experimental results for different methods on each dataset.

A. Toy Problems

In Fig. 2, a toy dataset is generated from three concentric circles, whose radiuses are $r_1 = 1$, $r_2 = 2$ and $r_3 = 3$, respectively. To verify the robustness of the methods, an experiment is carried out on the dataset with the noise level z ranging from 0.25 to 0.45, and the data points on each pattern are randomly sampled from $r_i - z$ to $r_i + z$, where $1 \le i \le 3$. Obviously, the closer z is to 0.5, the harder it is to achieve correct result. In the experiments, each data point from the middle circle is treated as the query, and the retrieval score for the top 80 ranks is calculated. The baseline is still obtained by L_2-norm. For each noise level, the re-ranking experiment is performed on 20 groups of randomly generated datasets, and then the average retrieval scores are evaluated.

To verify the effectiveness of the proposed hybrid regularization framework, we define another two fitting constraints: $\mathcal{F}_u^d$ and $\mathcal{F}_u^s$. And each of them contains one uniform regularization term: the square of L_2-norm. $\mathcal{F}_u^d$ associated with contextual dissimilarities $\boldsymbol{\varphi}_L$ is defined as

$$\begin{aligned}\mathcal{F}_u^d &= \frac{1}{2}\alpha\sum_{i=1}^{a}\frac{e_{ii}}{d_{ii}}\left(\varphi_{iL} - \frac{d_{ii}}{e_{ii}}\right)^2 + \frac{1}{2}\alpha\sum_{j=a+1}^{u}\left(\varphi_{jL} - \xi\right)^2 \\ &= \frac{1}{2}\alpha\left\|\mathbf{D}_A^{-\frac{1}{2}}\mathbf{E}_A^{\frac{1}{2}}\boldsymbol{\varphi}_{AL} - \mathbf{D}_A^{\frac{1}{2}}\mathbf{E}_A^{-\frac{1}{2}}\mathbf{1}\right\|_2^2 + \frac{1}{2}\alpha\left\|\boldsymbol{\varphi}_{BL} - \boldsymbol{\xi}_b\right\|_2^2,\end{aligned} \tag{32}$$

where $\boldsymbol{\varphi}_{AL} \in \mathbb{R}^a$, $\boldsymbol{\varphi}_{BL} \in \mathbb{R}^b$, $\boldsymbol{\varphi}_L = \left[\boldsymbol{\varphi}_{AL}^\top, \boldsymbol{\varphi}_{BL}^\top\right]^\top$, $\xi \gg \max_{i\in A}\frac{d_{ii}}{e_{ii}}$, and $\alpha > 0$ is the hyper-parameter. $\mathcal{F}_u^s$ associated with contextual similarities $\boldsymbol{\eta}_L$ is defined as

$$\begin{aligned}\mathcal{F}_u^s &= \frac{1}{2}\alpha\sum_{i=1}^{a}\frac{e_{ii}}{d_{ii}}\left(\eta_{iL} - \frac{e_{ii}}{d_{ii}}\right)^2 + \frac{1}{2}\alpha\sum_{j=a+1}^{u}\left(\eta_{jL}\right)^2 \\ &= \frac{1}{2}\alpha\left\|\mathbf{D}_A^{-\frac{1}{2}}\mathbf{E}_A^{\frac{1}{2}}\boldsymbol{\eta}_{AL} - \mathbf{D}_A^{-\frac{3}{2}}\mathbf{E}_A^{\frac{3}{2}}\mathbf{1}\right\|_2^2 + \frac{1}{2}\alpha\left\|\boldsymbol{\eta}_{BL}\right\|_2^2,\end{aligned} \tag{33}$$

where $\boldsymbol{\eta}_{AL} \in \mathbb{R}^a$, $\boldsymbol{\eta}_{BL} \in \mathbb{R}^b$, $\boldsymbol{\eta}_L = \left[\boldsymbol{\eta}_{AL}^\top, \boldsymbol{\eta}_{BL}^\top\right]^\top$, and $\alpha > 0$ is the hyper-parameter. Then, the corresponding regularization frameworks are defined as $\mathcal{J}_u^d\left(\boldsymbol{\varphi}_L\right) = \mathcal{S}_h\left(\boldsymbol{\varphi}_L\right) + \mathcal{F}_u^d\left(\boldsymbol{\varphi}_L\right)$ and $\mathcal{J}_u^s\left(\boldsymbol{\eta}_L\right) = \mathcal{S}_h\left(\boldsymbol{\eta}_L\right) + \mathcal{F}_u^s\left(\boldsymbol{\eta}_L\right)$. The idea of solving $\mathcal{J}_u^d$ and $\mathcal{J}_u^s$ is similar to the proposed algorithms, and the closed-form solutions are directly given as

$$\boldsymbol{\varphi}_L^* = \alpha f\left(\Lambda_u\right)\begin{bmatrix}\mathbf{1}_a \\ \boldsymbol{\xi}_b\end{bmatrix}, \tag{34}$$

and

$$\boldsymbol{\eta}_L^* = \alpha f\left(\Lambda_u\right)\mathbf{D}_U^{-2}\mathbf{E}_U^2\mathbf{1}, \tag{35}$$

where $\Lambda_u = \Lambda + \alpha\mathbf{H}_U\mathbf{L}_U$. Λ and $f(\cdot)$ are the same as those in Eq. 14, and $\mathbf{L}_U$ is a $u \times u$ diagonal matrix defined as

$$\mathbf{L}_U = \begin{bmatrix}\mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_B\end{bmatrix}, \tag{36}$$

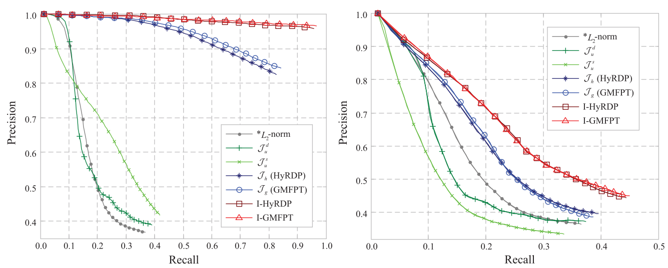
where $\mathbf{I}_B$ is a $b \times b$ identity matrix.

In this experiment, $\mathbf{W}$ is constructed by setting $K_d = K_s = 8$. Table I shows the average retrieval scores for different methods, and we further plot the P-R curves of the experiments with $z = 0.30$ and $z = 0.40$ in Fig. 5. As can be seen, HyRDP and GMFPT with the hybrid regularization framework perform much better than the methods with a uniform regularization term. Furthermore, I-HyRDP and I-GMFPT are superior to the others on most ranges of P-R curves, and obviously achieve the highest retrieval scores. With different noise levels $0.25 \le z \le 0.45$, the baselines are always dramatically improved by our methods, and the effects of noise can be greatly weakened and eliminated.

TABLE I

RETRIEVAL SCORES ON THE TOY DATASET. THE BASELINE IS MARKED WITH AN ASTERISK.

Method	Retrieval score				
	z=0.25	z=0.30	z=0.35	z=0.40	z=0.45
*L_2-norm	36.91%	36.82%	36.58%	36.55%	36.45%
$\mathcal{J}_u^d$	40.17%	38.95%	37.46%	37.36%	35.69%
$\mathcal{J}_u^s$	43.48%	41.84%	38.36%	33.56%	32.99%
$\mathcal{J}_h$ (HyRDP)	100.00%	82.52%	52.34%	39.57%	37.42%
$\mathcal{J}_g$ (GMFPT)	100.00%	84.30%	52.21%	38.63%	36.96%
I-HyRDP	100.00%	95.83%	71.94%	44.43%	40.05%
I-GMFPT	100.00%	96.66%	71.93%	44.98%	40.28%



(a) $z = 0.30$ (b) $z = 0.40$

Fig. 5. Quantitative results on the toy dataset with two noise levels. The baseline is marked with an asterisk.

B. Artificial Shape Retrieval

We test the proposed methods on two widely used shape datasets: Tari-1000 (TARI1k) dataset [42] and MPEG7 CE-Shape-1 part B (MPEG7) dataset [43]. In TARI1k, 1,000 shapes are divided into 50 categories, and each category contains 20 silhouettes. MPEG7 consists of 1,400 silhouettes divided into 70 categories. Each of them contains various objects, most of which are artificial targets. For each dataset, there also exist large variations in the objects from the same category, and it is a huge challenge for shape analysis.

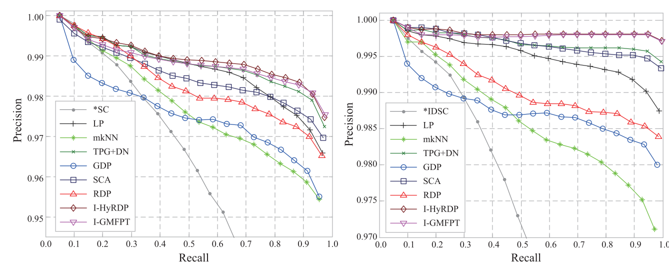
Two shape features, shape context (SC) [44] and inner distance shape context (IDSC) [45], are taken as the baselines in this experiment. For these two methods in the following experiments, 100 points are sampled from each shape contour to generate shape features, and a dynamic programming scheme is utilized for matching features. For TARI1k, the retrieval scores of SC and IDSC are 88.01% and 90.43%, respectively. And for MPEG7, SC and IDSC achieve the retrieval scores of 79.15% and 79.71%, respectively. It is notable that IDSC performs better than SC on these datasets because of lots of articulation changes within each class. Another thing to note is that, for MPEG7, the retrieval score is much stricter than the traditional bull's eye score, which saturates with many previous methods [8], [15], [26].

In the experiment on TARI1k, $\mathbf{W}$ is obtained by setting $K_d = K_s = 12$. Table II gives the retrieval scores obtained from the proposed methods and other approaches. And we further plot the P-R curves of these methods in Fig. 6. With IDSC selected as the baseline, our methods provide near-perfect retrieval scores: 99.72% for I-HyRDP and 99.71% for I-GMFPT. On the basis of SC, our approaches achieve

TABLE II

RETRIEVAL SCORES ON TARI1K. EACH BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH BASELINE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	Retrieval score	
	*SC [44]	*IDSC [45]
-	88.01%	90.43%
LP [4]	96.59%	98.75%
mkNN [32]	95.44%	97.11%
TPG+DN [23]	97.25%	99.43%
GDP [15]	95.52%	98.00%
SCA [19]	96.97%	99.34%
RDP [17]	96.53%	98.39%
I-HyRDP	97.47%	**99.72%**
I-GMFPT	**97.53%**	99.71%



(a) (b)

Fig. 6. Quantitative results on TARI1k with two baselines. Each baseline is marked with an asterisk.

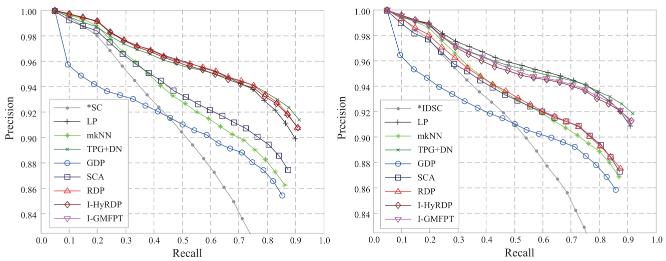
TABLE III

RETRIEVAL SCORES ON MPEG7. EACH BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH BASELINE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	Retrieval score	
	*SC [44]	*IDSC [45]
-	79.15%	79.71%
LP [4]	89.91%	90.86%
mkNN [32]	86.27%	86.90%
TPG+DN [23]	**91.40%**	**91.86%**
GDP [15]	85.86%	85.45%
SCA [19]	87.28%	87.44%
RDP [17]	86.65%	87.54%
I-HyRDP	90.79%	91.28%
I-GMFPT	90.78%	91.19%

97.47% for I-HyRDP and 97.53% for I-GMFPT. Specifically, in Fig. 6a and Fig. 6b, the previous methods such as SCA [19], TPG + DN [23], LP [4], have similar performance competing the proposed approaches and yet lower precision at high ranges of recall. The retrieval scores and the P-R curves demonstrate that our methods perform better than the others on TARI1k.

In the experiment on MPEG7, $\mathbf{W}$ is obtained by setting $K_d = K_s = 18$. The retrieval scores of the proposed methods and the competitors are shown in Table III. With SC as the baseline, I-HyRDP and I-GMFPT achieve the retrieval sores of 90.79% and 90.78%, respectively. With IDSC as the baseline, we have the retrieval sores of 91.28% for I-HyRDP



(a) (b)

Fig. 7. Quantitative results on MPEG7 with two baselines. Each baseline is marked with an asterisk.

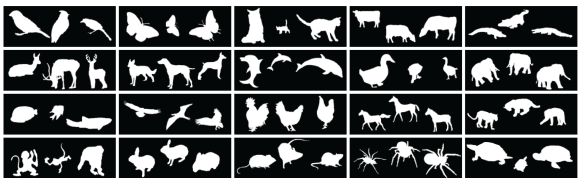


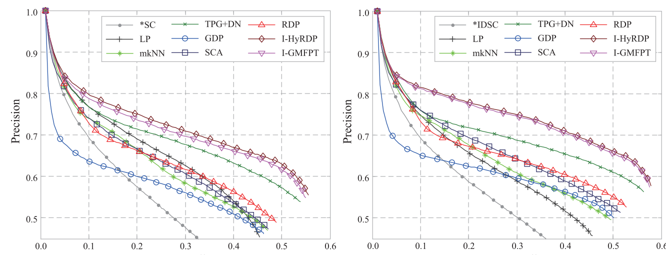
Fig. 8. Three typical images for each class from ANIM.

Fig. 9. Typical images of SWDLEAF.

TABLE IV

RETRIEVAL SCORES ON ANIM. EACH BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH BASELINE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	Retrieval score	
	*SC [44]	*IDSC [45]
-	38.38%	40.50%
LP [4]	45.44%	45.65%
mkNN [32]	47.09%	49.53%
TPG+DN [23]	53.86%	56.39%
GDP [15]	46.27%	50.09%
SCA [19]	47.26%	51.43%
RDP [17]	48.91%	52.70%
I-HyRDP	**55.51%**	**57.82%**
I-GMFPT	54.90%	57.51%



(a) (b)

Fig. 10. Quantitative results on ANIM with two baselines. Each baseline is marked with an asterisk.

and 91.19% for I-GMFPT. One can find that the proposed methods provide the second and the third best re-ranking performances that are close to the best results. As can be seen from the P-R curves in Fig. 7, our methods achieve comparable performances to TPG + DN [23] for most ranges of P-R curves, which are superior to the others.

C. Natural Shape Retrieval

We test our methods on two real datasets: Animal (ANIM) dataset [46] and Swedish leaf (SWDLEAF) dataset [47]. ANIM contains 2,000 manually extracted shapes divided into 20 categories. Since the silhouettes of living creatures contain various articulated and non-rigid deformations, it is a very challenging task for retrieval on this dataset. Fig. 8 gives the binary image examples of ANIM. SWDLEAF contains 1,125 leaves from 15 Swedish tree species. Because of high between-class similarities and large within-class deformations, the retrieval of leaves is fairly challenging. Fig. 9 shows typical image examples of SWDLEAF, one for each category.

In the previous works [47], [48], ANIM or SWDLEAF is adopted with training and testing splits, and employed only for classification. However, in this experiment, all the elements from each dataset are selected for testing the re-ranking performance of different algorithms. SC and IDSC are still selected as the shape matching methods, and the retrieval scores are calculated. For ANIM, The baseline results of SC and IDSC are 38.38% and 40.50%, respectively. For SWDLEAF, the baseline performances of SC and IDSC are 76.19% and 72.50%, respectively.

In the experiment on ANIM, $\mathbf{W}$ is constructed by setting $K_d = K_s = 20$. Table IV lists the re-ranking results of different methods, and the P-R curves are plotted in Fig. 10. Given IDSC as the baseline, the proposed algorithms achieve the highest precision values in almost the entire recall range, and provide the highest retrieval scores, 57.82% for I-HyRDP and 57.51% for I-GMFPT. Meanwhile, given SC as the baseline, I-HyRDP and I-GMFPT are also superior to the others, and we have the retrieval scores of 55.51% for I-HyRDP and 54.90% for I-GMFPT. Therefore, the retrieval scores of the proposed methods are outperforming the others, and the baselines are significantly improved by over 16%.

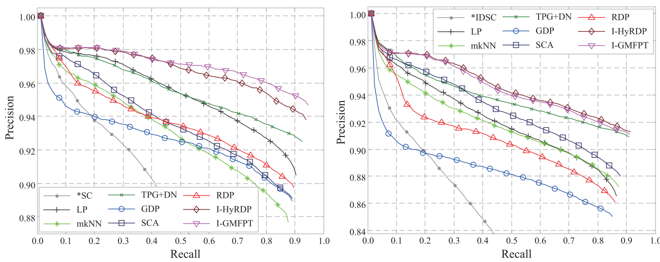
In the experiment on SWDLEAF, $\mathbf{W}$ is obtained by setting $K_d = K_s = 20$. The re-ranking results obtained by different methods are listed in Table V, and the P-R curves are demonstrated in Fig. 11. As can be seen, the P-R curves clearly show that I-HyRDP and I-GMFPT outperform the state-of-the-art algorithms. With SC selected as the baseline, the highest retrieval score of 94.68% is obtained by I-GMFPT, and I-HyRDP provides the second highest retrieval score of 93.83%. And based on IDSC, I-HyRDP and I-GMFPT achieve the retrieval scores of 91.28% and 91.11%, respectively. The proposed algorithms boost the baseline results of SC and IDSC by over 17% and 18%, respectively.

D. Natural Image Retrieval

In this subsection, the experiments on Ukbench (UKBENCH) image dataset [49] are carried out. This dataset

TABLE V
RETRIEVAL SCORES ON SWDLEAF. EACH BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH BASELINE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	Retrieval score	
	*SC [44]	*IDSC [45]
-	76.19%	72.50%
LP [4]	90.47%	86.55%
mkNN [32]	87.71%	87.22%
TPG+DN [23]	92.49%	90.89%
GDP [15]	88.96%	85.05%
SCA [19]	89.10%	88.00%
RDP [17]	89.73%	86.10%
I-HyRDP	93.83%	**91.28%**
I-GMFPT	**94.68%**	91.11%



(a) (b)

Fig. 11. Quantitative results on SWDLEAF with two baselines. Each baseline is marked with an asterisk.

contains 10,200 images from 2,550 different categories, and each category contains only 4 images. Instead of the retrieval score, the performance is evaluated by the average recall of the top 4 ranked images, referred as N-S score. In this experiment, three different features are selected as the baselines: Hamming Embedding (HE) [50], and HSV histogram (HSV) [51], and CNN based on Caffe framework [52]. Experimental results of the baseline methods on UKBENCH are already given in [51], and we directly introduce them into our re-ranking experiments.

In this experiment, $\mathbf{W}$ is constructed by setting $K_s = 4$, and λ is set to 2. The N-S scores of different methods on UKBENCH are listed in Table VI. As observed from Table VI, I-HyRDP and I-GMFPT achieve the highest N-S scores on the basis of HE and CNN. And, with HSV as the baseline, our algorithms provide the second and the third best scores that are close to the best performance achieved by SCA [19].

It is worth noting that the obtained N-S scores are not as good as some state-of-the-art results [17], [19], [53], [54]. However, as already emphasized, the proposed re-ranking method can not be treated as a stand-alone method, but a generic affinity learning approach, which is able to achieve better results on the basis of other baselines.

E. 3D Model Retrieval

ModelNet is a large-scale 3D CAD model dataset provided by Wu *et al.* [55]. A subset of this dataset named ModelNet40 is selected for the re-ranking experiment, and

TABLE VI
N-S SCORES ON UKBENCH. EACH BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH BASELINE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	N-S score		
	*HE [50]	*HSV [51]	*CNN [52]
-	3.573	3.195	3.397
LP [4]	3.529	2.443	3.355
mkNN [32]	3.693	3.236	3.507
GDP [15]	3.628	3.033	3.534
SCA [19]	3.711	**3.361**	3.580
RDP [17]	N/A	3.284	3.559
I-HyRDP	3.727	3.312	3.603
I-GMFPT	**3.730**	3.313	**3.617**

12,311 models are divided into 40 different categories. Similar to [55], we adopt the same testing dataset with 2,468 objects for evaluating the retrieval performance, and the number of 3D models from each category ranges from 20 to 100. Instead of traditional hand-crafted features, we take a deep 3D representation named 3D ShapeNets (3D-SN) [55] as the baseline. So, each 3D model is represented by the output of the 5th layer of 3D-SN, and then the input distances is further calculated by using L_2-norm.

In [55], every model from the testing dataset is rotated 30 degrees, and 12 poses for each one are obtained. Unlike the experiments in previous works, the model poses with the same rotation angle are selected to generate an independent dataset, and then re-ranking experiments are separately carried out on each dataset. As listed in Table VII, the baselines range from 50.03% to 51.57% in the 12 datasets.

In this experiment, $\mathbf{W}$ is obtained with $\gamma = 1$ and $K_d = K_s = 25$. For the proposed model of HyRDP, we set $\alpha = \beta = 100$. As can be seen from Table VII, in almost all cases, the proposed I-HyRDP outperforms the other methods, and I-GMFPT is also listed in the top 3. We further plot the percentage of correct results among the first K most similar objects in Fig. 12, where $K = 1, \cdots, 100$. It can be observed that I-HyRDP obviously provides the highest precision values in almost the entire range of K. Meanwhile, I-GMFPT is superior to or at least comparable with the competitors.

Similar to the experiment on UKBENCH, the obtained re-ranking results are still not as good as some state-of-the-art results [56], [57]. However, as mentioned before, we mainly focus on the improvement on the selected baseline in this experiment rather than the retrieval score itself.

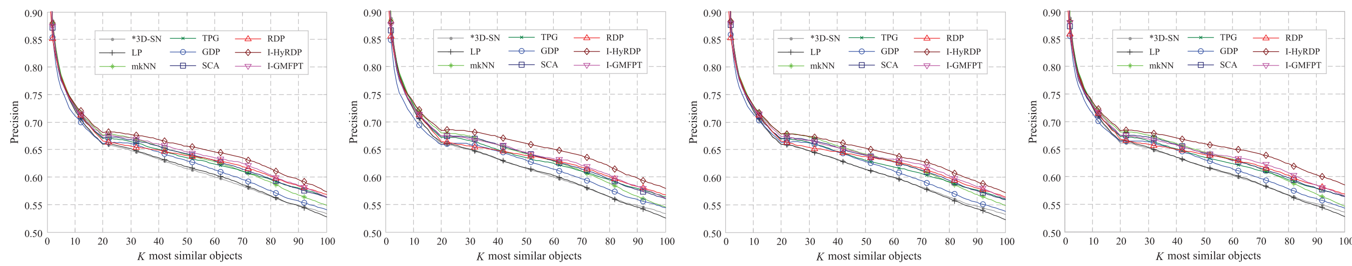
F. The Choice of Parameters

The parameters in the proposed algorithms are divided into three groups according to the roles. First, γ, K_d and K_s are used to determine the weighted matrix $\mathbf{W}$. Second, for the proposed HyRDP, α and β are applied to keep the balance among the three terms of the hybrid regularization framework. At last, λ plays a direct role in the iterative re-ranking process. So, we discuss these three groups separately.

TABLE VII

RETRIEVAL SCORES ON MODELNET40. THE BASELINE IS MARKED WITH AN ASTERISK, AND THE THREE BEST RESULTS FOR EACH POSE ARE MARKED WITH BOLDFACE BLUE, GREEN, AND ORANGE FONTS, RESPECTIVELY.

Method	Pose1	Pose2	Pose3	Pose4	Pose5	Pose6	Pose7	Pose8	Pose9	Pose10	Pose11	Pose12
*3D-SN [55]	50.13%	51.02%	51.30%	50.08%	51.28%	51.50%	50.03%	51.09%	51.49%	50.38%	51.42%	51.57%
LP [4]	50.22%	51.16%	51.45%	50.12%	51.08%	51.42%	49.88%	50.91%	51.56%	50.35%	51.30%	51.68%
mkNN [32]	51.85%	53.07%	53.21%	51.57%	53.09%	53.21%	51.91%	53.16%	53.44%	51.73%	53.33%	53.43%
TPG+DN [23]	53.37%	54.27%	55.01%	53.29%	54.48%	55.28%	53.05%	54.07%	55.11%	53.35%	54.62%	55.57%
GDP [15]	50.82%	52.27%	53.07%	50.95%	52.28%	52.92%	50.63%	52.41%	53.06%	50.90%	52.44%	53.23%
SCA [19]	52.93%	54.54%	54.86%	52.73%	54.62%	54.81%	52.56%	54.27%	54.85%	52.96%	54.81%	55.01%
RDP [17]	53.27%	54.52%	54.88%	53.22%	54.49%	54.89%	52.92%	54.50%	54.80%	53.36%	54.66%	55.05%
I-HyRDP	**54.34%**	**55.90%**	**56.46%**	**54.99%**	**55.41%**	**56.17%**	**54.28%**	54.78%	**56.03%**	**55.39%**	**55.56%**	**56.34%**
I-GMFPT	53.59%	54.84%	54.81%	53.39%	54.17%	54.88%	53.36%	**54.96%**	55.06%	53.75%	54.91%	56.17%



(a) Pose1 (b) Pose4 (c) Pose7 (d) Pose10

Fig. 12. Quantitative results on ModelNet40 in four poses. The baseline is marked with an asterisk.

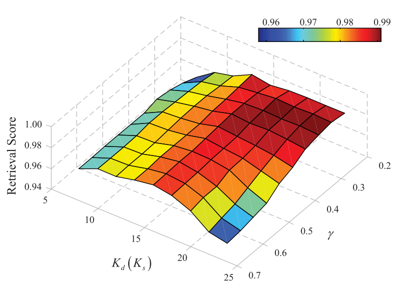
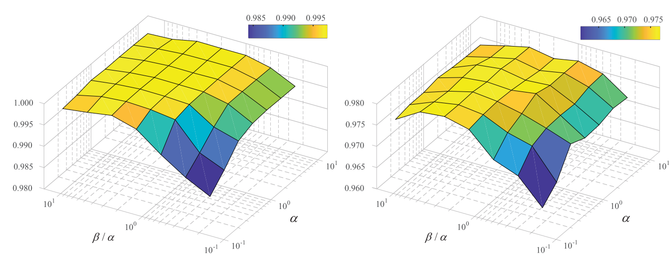


Fig. 13. Impacts of γ and $K_d\,(K_s)$ on re-ranking performance on TARIk-SUB. Retrieval score of I-GMFPT based on SC is present.



(a) I-HyRDP based on IDSC (b) I-HyRDP based on SC

Fig. 14. Impacts of α and β/α on retrieval score on SWDLEAF-SUB1.

Firstly, we construct a testing dataset named TARI1k-SUB with 525 images randomly selected from TARI1k. It contains 35 classes with 15 different objects in each class ($35 \times 15 = 525$). Given SC as the baseline, Fig. 13 shows the performance of I-GMFPT with $\lambda = 5$, and our method achieves stable results with γ close to 0.4 and $K_d\,(K_s)$ around 12. So, we set $\gamma = 0.4$ in our experiments unless specified otherwise. And K_d and K_s are set to the same value empirically for convenience, and determined by small-scale experiments. It is worth noting that, the construction of $\mathbf{W}$ is essential for diffusion process, and similar parameters have also been applied to many other variants [4], [6], [9], [26], [27].

Secondly, we build a testing dataset named SWDLEAF-SUB1 with 525 images randomly selected from SWDLEAF. It contains 15 classes with 35 different objects in each class ($15 \times 35 = 525$). In the experiments, $\mathbf{W}$ is constructed by setting $K_d = K_s = 12$. As shown in Fig. 14, α and β/α are treated as the variables, and we have the performances of I-HyRDP on the basis of IDSC and SC respectively. From Fig. 14, the stable results can be obtained with $0 < \alpha \leq 10$ and $\alpha \leq \beta \leq 10\alpha$. In our experiments, we set $\alpha = 5$ and $\beta = 5\alpha = 25$ unless specified otherwise.

Finally, we discuss the impacts of λ. We construct another testing dataset named SWDLEAF-SUB2 with 525 images randomly selected from SWDLEAF, and it also contains 15 classes each having 35 images ($15 \times 35 = 525$). Fig. 15 shows the performances of SC + I-HyRDP and IDSC + I-GMFPT on SWDLEAF-SUB1 and SWDLEAF-SUB2, where λ ranges from 1 to 35. As can be seen from Fig. 15, our methods provide reliable retrieval scores with λ close to a small value. And λ is set to 5 in our experiments unless specified otherwise.

It is worth reminding that the three parameters, γ, α and β, are closely related to the baseline. So, for different measures, the choice of them can be determined by small-scale experiments.

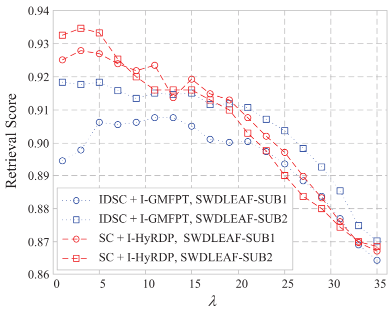


Fig. 15. Impact of λ on retrieval score on testing datasets.

G. Time Complexity Analysis

Obviously, the post-processing procedure is independent from the process of calculating input distances, and we are concerned only with the time complexity of the re-ranking algorithm. In our approach, the cost for each re-ranking iteration is mainly determined by the computation of $\boldsymbol{\mu}_{L^t}$ or $\boldsymbol{\psi}_{L^t}$. We have $O\left(u_t^3\right)$ for the closed-form solution and $O\left(m_t u_t^2\right)$ for the iterative solution, where $u_t = \#U^t$ and $u_t \ll n$. m_t denotes the iteration number of the iterative solution in the iteration t of re-ranking process, and mostly varies from 10^2 to 10^3 in our experiments. In practice, since many elements from u^t belong to the neighborhood of each other, we have $n_t \ll t\lambda K_s^2$, and n_t is near to or less than 10^2 for most of the time in our experiment. Specifically, our methods are tested on a common personal computer with a 4.00GHz CPU and 16GB of physical RAM. Then, the average time of one re-ranking is less than 0.02s on SWDLEAF, and we have the average time close to 0.3s on ANIM.

VI. Conclusion

In this article, as a novel variant of diffusion process, the hybrid regularization framework named HyRDP is proposed for visual re-ranking on the basis of a two-part fitting constraint. And GMFPT as a general form of MFPT is also introduced as the contextual dissimilarity, which denotes the mean time-steps for the state transition. By analysis of the mechanism behind GMFPT, it is find that the objective function behind GMFPT is basically the same as that of HyRDP. Moreover, we further present an iterative re-ranking process on the basis of the proposed contextual dissimilarity, and the underlying manifold structure can be effectively obtained with the similar objects gradually searched and explicitly labeled. The experiments on different databases demonstrate that baseline retrieval results can be greatly improved by our methods.

For the same retrieval task, different baselines may tackle the problem from different angles, and thus often provide complementary information. Many variants of diffusion process have already been used to fuse multiple metrics [25], [29]. In this sense, the proposed algorithms are not limited to one individual type of similarity but can also be applied to similarity fusion, which will be part of our future work. Meanwhile, in recent years, cross-modal retrieval has garnered particular attention because of the explosive growth of multimedia data from various search engines and social media [58], [59]. And the proposed methods also have great potential to be applied to this task.

Appendix A
Proposition 2

$\mathbf{W}_{UU}$ and $\mathbf{H}_U$ are given in Subsection III-A. $\mathbf{R}_U$ is defined as a $u \times u$ diagonal matrix with $r_{ii} > 0$ and $r_{jj} \geq 0$, where $1 \leq i \leq a$ and $a+1 \leq j \leq u$. Then, we have $\rho(\mathbf{Q}) < 1$, where $\mathbf{Q} = (\mathbf{H}_U + \mathbf{R}_U)^{-1}\mathbf{W}_{UU}$.

Proof: Given $m = u+1$, we construct $\mathbf{W}' = \left[w'_{ij}\right]_{m\times m}$ as follows

$$\mathbf{W}' = \begin{bmatrix} \mathbf{W}_{UU} & \boldsymbol{r} \\ \boldsymbol{r}^\top & 0 \end{bmatrix}, \tag{37}$$

where $\boldsymbol{r} = [r_{11}, r_{22}, \cdots, r_{uu}]^\top$. We construct a weighted graph $\mathcal{G}'$ based on S' and $\mathbf{W}'$, where $S' = U + \{m\}$. In $\mathcal{G}'$, we certainly have $i \sim m$ since $w'_{im} > 0$ for $\forall i \in A$. The corresponding transition matrix is $\mathbf{P}' = \mathbf{D}'^{-1}\mathbf{W}' = \left[p'_{ij}\right]_{m\times m}$, where the $m \times m$ diagonal matrix $\mathbf{D}'$ can be expressed as

$$\mathbf{D}' = \begin{bmatrix} \mathbf{H}_U + \mathbf{R}_U & \mathbf{0} \\ \mathbf{0} & \sum_{i=1}^{u} r_{ii} \end{bmatrix}. \tag{38}$$

In $\mathcal{G}$, $\exists k \in L$ satisfies $i \sim k$ according to Subsection III-A, where $i \in B$. Since there is no direct connection from i to any node in L, $\mathcal{G}$ contains a path from i to k through $j \in A$, and it does not pass any node in L before reaching k. With U taken as the vertices, the subgraph of $\mathcal{G}'$ is identical to that of $\mathcal{G}$. So we have the partial path from i to j in $\mathcal{G}'$, the same as that in $\mathcal{G}$. Then, we have $i \sim A$ and $i \sim m$ for $\forall i \in B$ in $\mathcal{G}'$.

Therefore, $\mathcal{G}'$ is a strongly connected graph, and we have $p'^{(k)}_{ij} = \Pr(X_k = j | X_0 = i) > 0$ for a large enough k. Let q_{ij} denote the entry of $(\mathbf{Q})^k$, and we obtain

$$\sum_{j=1}^{u} q_{ij} \leq \sum_{j=1}^{u} p'^{(k)}_{ij} = 1 - p'^{(k)}_{im} < 1. \tag{39}$$

Therefore, $\left\|(\mathbf{Q})^k\right\|_\infty = \max_{1\leq i\leq u} \sum_{j=1}^{u} \left|q_{ij}\right| < 1$, and we obtain $\rho(\mathbf{Q}) \leq \left\|(\mathbf{Q})^k\right\|_\infty^{\frac{1}{k}} < 1$.

Appendix B
Derivation Process Of Eq. 19

We define $f_{iL}^{(k)}$ as a conditional probability as

$$f_{iL}^{(k)} = \Pr\left(X_k \in L, X_m \notin L, 1 \leq m \leq k-1 | X_0 = i\right), \tag{40}$$

where $i \in S$. Then, we have $f_{iL}^{(k)} = \sum_{j=1}^{u} p_{ij} f_{jL}^{(k-1)}$ and $f_{iL}^{(1)} = \sum_{j=u+1}^{n} p_{ij}$, which can be expressed as

$$\boldsymbol{f}_{nL}^{(k)} = \begin{bmatrix} \mathbf{P}_{UU} \\ \mathbf{P}_{LU} \end{bmatrix} \boldsymbol{f}_{uL}^{(k-1)}, \tag{41}$$

and

$$\boldsymbol{f}_{nL}^{(1)} = \begin{bmatrix} \mathbf{P}_{UL} \\ \mathbf{P}_{LL} \end{bmatrix} \mathbf{1}. \tag{42}$$

where $\boldsymbol{f}_{jL}^{(i)} = \left[f_{1L}^{(i)}, \cdots, f_{jL}^{(i)} \right]^{\top}$. According to the definition of μ_{iL}, we have $\mu_{iL} = \sum_{j \notin L} p_{ij} \mu_{jL} + \sum_{k=1}^{\infty} f_{iL}^{(k)}$, and it can be written as

$$\left(\mathbf{I} - \begin{bmatrix} \mathbf{P}_{UU} & \mathbf{0} \\ \mathbf{P}_{LU} & \mathbf{0} \end{bmatrix} \right) \boldsymbol{\mu}_{SL} = \sum_{k=1}^{\infty} \boldsymbol{f}_{nL}^{(k)}. \tag{43}$$

According to Eq. 41 and Eq. 42, we have

$$\sum_{k=1}^{\infty} \boldsymbol{f}_{nL}^{(k)} = \begin{bmatrix} \mathbf{P}_{UU} \\ \mathbf{P}_{LU} \end{bmatrix} \sum_{k=0}^{\infty} (\mathbf{P}_{UU})^k \mathbf{P}_{UL} \mathbf{1} + \begin{bmatrix} \mathbf{P}_{UL} \\ \mathbf{P}_{LL} \end{bmatrix} \mathbf{1}. \tag{44}$$

As $\mathbf{P}_{UU} = (\mathbf{H}_U + \mathbf{E}_U)^{-1} \mathbf{W}_{UU}$, we have $\rho(\mathbf{P}_{UU}) < 1$ according to APPENDIX A, and $\sum_{k=0}^{\infty} (\mathbf{P}_{UU})^k \mathbf{P}_{UL} \mathbf{1} = \mathbf{1}$, where $\sum_{k=0}^{\infty} (\mathbf{P}_{UU})^k = (\mathbf{I} - \mathbf{P}_{UU})^{-1}$. Thus, we have $\sum_{k=1}^{\infty} f_{iL}^{(k)} = 1$ for $1 \leq i \leq n$. In Eq. 43, the coefficient matrix in the left-hand side is invertible, since the main diagonal blocks are invertible. Finally, Eq. 19 can be derived from Eq. 43.

References

[1] D. Zhou, J. Weston, A. Gretton, O. Bousquet, and B. Schölkopf, "Ranking on data manifolds," in *Proc. Adv. Neural Inf. Process. Syst.*, 2004, pp. 169–176.

[2] S. Bai, X. Bai, Q. Tian, and L. J. Latecki, "Regularized diffusion process for visual retrieval," in *Proc. AAAI Conf. Artif. Intell.*, 2017, pp. 3967–3973.

[3] H. Jegou, C. Schmid, H. Harzallah, and J. Verbeek, "Accurate image search using the contextual dissimilarity measure," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 1, pp. 2–11, Jan. 2010.

[4] X. Bai, X. Yang, L. J. Latecki, W. Liu, and Z. Tu, "Learning context-sensitive shape similarity by graph transduction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 5, pp. 861–874, May 2010.

[5] A. Egozi, Y. Keller, and H. Guterman, "Improving shape retrieval by spectral matching and meta similarity," *IEEE Trans. Image Process.*, vol. 19, no. 5, pp. 1319–1327, May 2010.

[6] J. Wang, Y. Li, X. Bai, Y. Zhang, C. Wang, and N. Tang, "Learning context-sensitive similarity by shortest path propagation," *Pattern Recognit.*, vol. 44, nos. 10–11, pp. 2367–2374, Oct. 2011.

[7] X. Yang, L. Prasad, and L. J. Latecki, "Affinity learning with diffusion on tensor product graph," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 28–38, Jan. 2013.

[8] S. Bai, S. Sun, X. Bai, Z. Zhang, and Q. Tian, "Smooth neighborhood structure mining on multiple affinity graphs with applications to context-sensitive similarity," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 592–608.

[9] Y. Zhou, X. Bai, W. Liu, and L. J. Latecki, "Similarity fusion for visual tracking," *Int. J. Comput. Vis.*, vol. 118, no. 3, pp. 337–363, Jul. 2016.

[10] S. Bai, X. Bai, and Q. Tian, "Scalable person re-identification on supervised smoothed manifold," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3356–3365.

[11] S. Bai, P. Tang, P. H. S. Torr, and L. J. Latecki, "Re-ranking via metric fusion for object retrieval and person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 740–749.

[12] H. Li, H. Lu, Z. Lin, X. Shen, and B. Price, "Inner and inter label propagation: Salient object detection in the wild," *IEEE Trans. Image Process.*, vol. 24, no. 10, pp. 3176–3186, Oct. 2015.

[13] T. Tong, K. Gray, Q. Gao, L. Chen, and D. Rueckert, "Nonlinear graph fusion for multi-modal classification of Alzheimer's disease," in *Proc. Int. Workshop Mach. Learn. Med. Imag.*, 2015, pp. 77–84.

[14] J. Almeida, D. C. G. Pedronette, B. C. Alberton, L. P. C. Morellato, and R. D. S. Torres, "Unsupervised distance learning for plant species identification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5325–5338, Dec. 2016.

[15] M. Donoser and H. Bischof, "Diffusion processes for retrieval revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 1320–1327.

[16] D. Zhou, O. Bousquet, T. N. Lal, J. Weston, and B. S. Olkopf, "Learning with local and global consistency," in *Proc. Adv. Neural Inf. Process. Syst.*, 2004, pp. 321–328.

[17] S. Bai, X. Bai, Q. Tian, and L. J. Latecki, "Regularized diffusion process on bidirectional context for object retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 5, pp. 1213–1226, May 2019.

[18] C. Deng, R. Ji, W. Liu, D. Tao, and X. Gao, "Visual reranking through weakly supervised multi-graph learning," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2600–2607.

[19] S. Bai and X. Bai, "Sparse contextual activation for efficient visual re-ranking," *IEEE Trans. Image Process.*, vol. 25, no. 3, pp. 1056–1069, Mar. 2016.

[20] X. Yang, X. Bai, and L. J. Latecki, "Improving shape retrieval by learning graph transduction," in *Proc. Eur. Conf. Comput. Vis.*, vol. 5305, 2008, pp. 788–801.

[21] X. Yang, S. Koknar-Tezel, and L. J. Latecki, "Locally constrained diffusion process on locally densified distance spaces with applications to shape retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 357–364.

[22] J. Jiang, B. Wang, and Z. Tu, "Unsupervised metric learning by self-smoothing operator," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 794–801.

[23] X. Yang and L. J. Latecki, "Affinity learning on a tensor product graph with applications to shape and image retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, vol. 42, no. 7, Jun. 2011, pp. 2369–2376.

[24] X. Bai, B. Wang, X. Wang, W. Liu, and Z. Tu, "Co-transduction for shape retrieval," in *Proc. Eur. Conf. Comput. Vis.*, vol. 6313, 2010, pp. 328–341.

[25] X. Bai, B. Wang, C. Yao, W. Liu, and Z. Tu, "Co-transduction for shape retrieval," *IEEE Trans. Image Process.*, vol. 21, no. 5, pp. 2747–2757, May 2012.

[26] B. Wang, J. Jiang, W. Wang, Z.-H. Zhou, and Z. Tu, "Unsupervised metric fusion by cross diffusion," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2997–3004.

[27] L. Luo, C. Shen, C. Zhang, and A. van den Hengel, "Shape similarity analysis by self-tuning locally constrained mixed-diffusion," *IEEE Trans. Multimedia*, vol. 15, no. 5, pp. 1174–1183, Aug. 2013.

[28] S. Bai, Z. Zhou, J. Wang, X. Bai, L. J. Latecki, and Q. Tian, "Ensemble diffusion for retrieval," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 774–783.

[29] S. Bai, Z. Zhou, J. Wang, X. Bai, L. J. Latecki, and Q. Tian, "Automatic ensemble diffusion for 3D shape and image retrieval," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 88–101, Jan. 2019.

[30] D. C. G. Pedronette and R. da S. Torres, "Image re-ranking and rank aggregation based on similarity of ranked lists," *Pattern Recognit.*, vol. 46, no. 8, pp. 2350–2360, Aug. 2013.

[31] X. Shen, Z. Lin, J. Brandt, S. Avidan, and Y. Wu, "Object retrieval and localization with spatially-constrained similarity measure and k-NN re-ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3013–3020.

[32] P. Kontschieder, M. Donoser, and H. Bischof, "Beyond pairwise shape similarity analysis," in *Proc. Asian Conf. Comput. Vis.*, vol. 5996, 2009, pp. 655–666.

[33] D. C. Guimarães Pedronette, O. A. B. Penatti, and R. da S. Torres, "Unsupervised manifold learning using reciprocal kNN graphs in image re-ranking and rank aggregation tasks," *Image Vis. Comput.*, vol. 32, no. 2, pp. 120–130, Feb. 2014.

[34] S. Redner, *A Guide to First-Passage Processes*. Cambridge, U.K.: Cambridge Univ. Press, 2001.

[35] S. Condamin, O. Bénichou, V. Tejedor, R. Voituriez, and J. Klafter, "First-passage times in complex scale-invariant media," *Nature*, vol. 450, no. 7166, pp. 77–80, Nov. 2007.

[36] B. Jiang, L. Zhang, H. Lu, C. Yang, and M.-H. Yang, "Saliency detection via absorbing Markov chain," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 1665–1672.

[37] J. Sun, H. Lu, and X. Liu, "Saliency region detection based on Markov absorption probabilities," *IEEE Trans. Image Process.*, vol. 24, no. 5, pp. 1639–1649, May 2015.

[38] D. Zheng, W. Liu, and H. Wang, "Improving shape retrieval by fusing generalized mean first-passage time," in *Proc. Int. Conf. Neural Inf. Process.*, 2017, pp. 439–448.

[39] R. Fagin and L. Stockmeyer, "Relaxing the triangle inequality in pattern matching," *Int. J. Comput. Vis.*, vol. 30, no. 3, pp. 219–231, 1998.

[40] R. C. Veltkamp, "Shape matching: Similarity measures and algorithms," in *Proc. Int. Conf. Shape Modeling Appl.*, 2001, pp. 188–197.

[41] M. Szummer and T. Jaakkola, "Partially labeled classification with Markov random walks," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2001, pp. 945–952.

[42] E. Baseski, A. Erdem, and S. Tari, "Dissimilarity between two skeletal trees in a context," *Pattern Recognit.*, vol. 42, no. 3, pp. 370–385, Mar. 2009.

[43] L. J. Latecki, R. Lakamper, and T. Eckhardt, "Shape descriptors for non-rigid shapes with a single closed contour," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2000, pp. 424–429.

[44] S. Belongie, J. Malik, and J. Puzicha, "Shape matching and object recognition using shape contexts," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 4, pp. 509–522, Apr. 2002.

[45] H. Ling and D. W. Jacobs, "Shape classification using the inner-distance," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 2, pp. 286–299, Feb. 2007.

[46] X. Bai, W. Liu, and Z. Tu, "Integrating contour and skeleton for shape classification," in *Proc. IEEE 12th Int. Conf. Comput. Vis. Workshops, ICCV Workshops*, Sep. 2009, pp. 360–367.

[47] O. J. O. Söderkvist, "Computer vision classification of leaves from Swedish trees," M.S. thesis, Linköping Univ., Linköping, Sweden, Sep. 2001.

[48] X. Bai, C. Rao, and X. Wang, "Shape vocabulary: A robust and efficient shape representation for shape matching," *IEEE Trans. Image Process.*, vol. 23, no. 9, pp. 3935–3949, Sep. 2014.

[49] D. Nister and H. Stewenius, "Scalable recognition with a vocabulary tree," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2, Jun. 2006, pp. 2161–2168.

[50] H. Jegou, M. Douze, and C. Schmid, "Hamming embedding and weak geometric consistency for large scale image search," in *Proc. Eur. Conf. Comput. Vis.*, 2008, pp. 304–317.

[51] L. Zheng, S. Wang, L. Tian, F. He, Z. Liu, and Q. Tian, "Query-adaptive late fusion for image search and person re-identification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1741–1750.

[52] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," 2014, *arXiv:1408.5093*. [Online]. Available: http://arxiv.org/abs/1408.5093

[53] L. Zheng, S. Wang, J. Wang, and Q. Tian, "Accurate image search with multi-scale contextual evidences," *Int. J. Comput. Vis.*, vol. 120, no. 1, pp. 1–13, Oct. 2016.

[54] A. Gordo, J. Almazán, J. Revaud, and D. Larlus, "End-to-end learning of deep visual representations for image retrieval," *Int. J. Comput. Vis.*, vol. 124, no. 2, pp. 237–254, Sep. 2017.

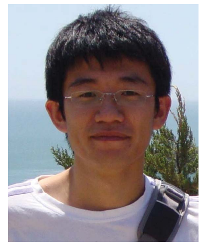
[55] Z. Wu *et al.*, "3D ShapeNets: A deep representation for volumetric shapes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1912–1920.

[56] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3D shape recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 945–953.

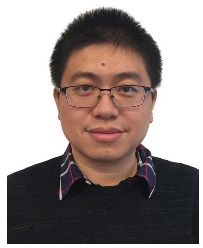
[57] S. Bai, X. Bai, Z. Zhou, Z. Zhang, and L. J. Latecki, "GIFT: A real-time and scalable 3D shape search engine," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5023–5032.

[58] C. Deng, Z. Chen, X. Liu, X. Gao, and D. Tao, "Triplet-based deep hashing network for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 3893–3903, Aug. 2018.

[59] C. Li, C. Deng, N. Li, W. Liu, X. Gao, and D. Tao, "Self-supervised adversarial hashing networks for cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4242–4251.



Danchen Zheng (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the Faculty of Electronic Information Electrical Engineering, Dalian University of Technology, Dalian, China, in 2006, 2009, and 2014, respectively. He was a Postdoctoral Fellow, jointly cultivated by the Dalian University of Technology, from 2014 to 2018. He is currently an Associate Professor with the Faculty of Electronic Information Electrical Engineering, Dalian University of Technology. His research interests include image recognition and retrieval, shape analysis, and information systems.



Jianchao Fan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Control Science and Engineering, Dalian University of Technology, Dalian, China, in 2007 and 2012, respectively. He was a Postdoctoral Fellow, jointly cultivated by The Chinese University of Hong Kong and the Dalian University of Technology, from 2013 to 2016, and also a Visiting Scholar with the Department of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, USA, from 2018 to 2019. He has been an Professor with the Department of Ocean Remote Sensing, National Marine Environmental Monitoring Center, Dalian, since 2019. He has authored or coauthored two books, more than 70 scientific articles in international journals and conference proceedings, such as the IEEE TRANSACTIONS ON FUZZY SYSTEMS, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Pattern Recognition*. His current research interests include marine SAR image processing, polarimetric SAR, deep learning neural networks, and heuristic swarm intelligent optimization.

Min Han (Senior Member, IEEE) received the B.S. and M.S. degrees from the Department of Electrical Engineering, Dalian University of Technology, Dalian, China, in 1982 and 1993, respectively, and the M.S. and Ph.D. degrees from Kyushu University, Fukuoka, Japan, in 1996 and 1999, respectively. Since 2003, she has been a Professor with the Faculty of Electronic Information and Electrical Engineering, Dalian University of Technology. She serves as the Deputy Director for the Chinese Society of Instrumentation Youth Work Committee, the Director for the institute of Liaoning Province System Simulation, a Committee Member for the Chinese Society of Artificial Intelligence, a Consultant of the Jiangsu Province Department of Science and Technology, the Deputy Director of the Institute of Fuzzy Information Processing and Machine Intelligence, Dalian University of Technology, and an Organizing Chair of ISNN2013, ICICIP2014, and ICIST2016. She has visited to Washington University in St. Louis, USA, in 2009. She has authored five books and more than 300 articles in international journals and conference proceedings. Her current research interests include complex system modeling and forecasting method, time series analysis and forecasting, artificial intelligence method, neural networks and chaos, and their applications to control and identification.